

GEOPHYSICS introduces new section on multiphysics and joint inversion

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Geophysicists seek to extract useful and potentially actionable information about the subsurface by interpreting various types of geophysical data together with prior geologic information. It is well recognized that reliable imaging, characterization, and monitoring of subsurface systems require integration of multiple sources of information from a multitude of geoscientific data sets. With increasing data volumes and computational power, new data types, constant development of inversion algorithms, and the advent of the big data era, GEOPHYSICS editors see multiphysics integration as an effective means of meeting some of the challenges arising from imaging subsurface systems with higher resolution and reliability as well as exploring geologically more complicated areas. To advance the field of multiphysics integration and to showcase its added value, GEOPHYSICS will introduce a new section “Multiphysics and Joint Inversion” in 2021. Submissions are accepted now.

Definition of data integration, joint inversion, and multiphysics

Different geophysical methods have gone through decades of independent development and advancement. Such separations occurred by necessity as our field and profession sought to advance and refine each subdiscipline. The parallel developments speak to the need of integrating different methods and complementary information from different data sets. Much of our profession has involved making integrated use of multiple methods on a routine basis. A multiphysics approach is the natural and logical next stage in our field as we seek to recombine and reconfigure these well-developed methods and their quantitative interpretation tools to increase reliability and resolution of subsurface imaging.

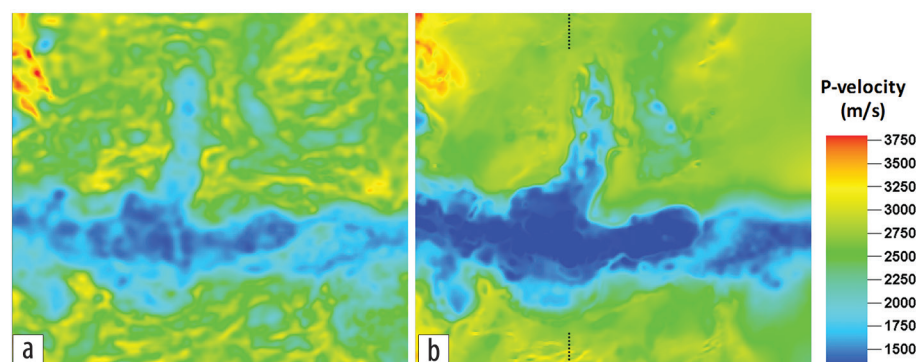


Figure 1. P-wave velocity distributions at 340 m elevation (mean sea level) derived from (a) standalone tomography and (b) joint inversion with resistivity as the reference model through cross-gradient regularization, which shows improved continuity and amplitudes. Courtesy of Colombo et al. (2016).

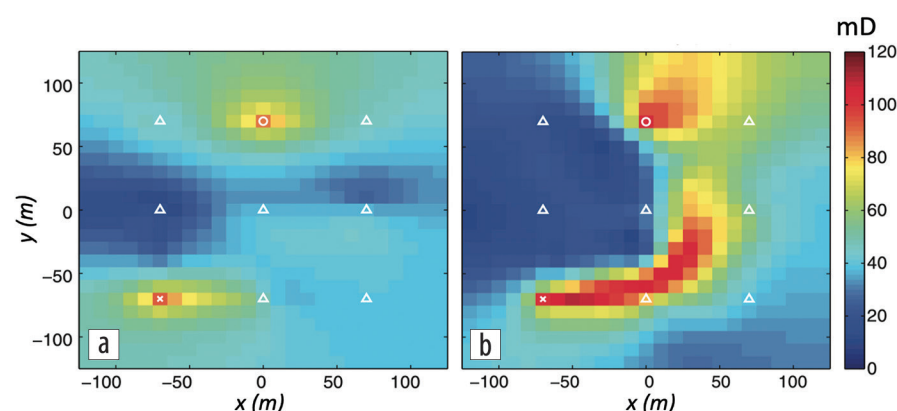


Figure 2. (a) Permeability in a synthetic reservoir recovered from the inversion of production data alone. (b) Multiphysics inversion of time-lapse electromagnetic, seismic, and production data. The multiphysics inversion recovers the correct spatial pattern and yields better-defined spatial continuity and reliability with the correct peak values. Modified from Liang et al. (2016).

By multiphysics, we refer to the collection of formalisms of integrated interpretation of multiple geophysical and geoscientific data sets for the purpose of imaging and characterizing subsurface systems that are the objects of geophysical investigation. The essence of multiphysics integration is to use and combine complementary information from different geophysical, geologic, petrophysical, and/or production data sets. The objective is the creation of new knowledge. By this definition, two or more geophysical data sets and/or additional geoscientific data are

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involved in a systematic manner. Examples of such approaches include direct integration of different data sets to image subsurface structures, integrated interpretation of multiple inversions using a generally applicable formulation, sequential and joint inversions of multiple data sets, statistical and machine learning-driven information extraction from multiple data sets, or a hybrid of these methods. The combination of various integration methods opens the avenue to a new branch of geophysical interpretation consisting of the science of data integration and information extraction. The ultimate goal is to produce subsurface images and models that yield more information and have higher resolution and lower uncertainty.

Examples

A plethora of publications exists on multiphysics integration, ranging from direct data integration, joint inversion methodologies (Gallardo and Meju, 2004; Moorkamp et al., 2011; Sun and Li, 2016; Astic and Oldenburg, 2019), application of such methodologies in near-surface velocity imaging (Colombo et al., 2016), integrated interpretation and joint inversion in mineral exploration (Lelièvre et al., 2012; Kamm et al., 2015; Melo et al., 2017), reservoir characterization (Hoversten et al., 2006; Du and MacGregor, 2010), and coupled simulation and inversions of time-lapse geophysical data and production data (Dadashpour et al., 2008; Glegola et al., 2012; Liang et al., 2016).

We show two examples of multiphysics integration and hope that these illustrate the types of submissions sought for the new GEOPHYSICS section. The first example (Figure 1) is the integration of seismic and helicopter-borne transient electromagnetic data through simultaneous joint inversion that significantly improves characterization of the near-surface velocity variation in a structure-controlled wadi (Colombo et al., 2016). The second example (Figure 2) is the coupled inversion of modeled time-lapse electromagnetic and seismic data with injection and production data for improved reservoir modeling (Liang et al., 2016). The multiphysics inversion result yields a significantly improved permeability model compared with that from the inversion of production data alone.

Expected submissions

We welcome submissions of research work on both theoretical/algorithmic developments and case studies with, but not limited to, the following applications:

- Oil and gas exploration
- Mineral exploration
- Geothermal exploration
- CO₂ storage and monitoring
- Reservoir characterization and monitoring
- Unconventional reservoirs
- Groundwater hydrogeophysics
- Velocity model building
- Near-surface modeling

The submissions are expected to elaborate on how integration of multiphysics data sets is achieved and what new knowledge and information results from such an integration. **TLE**

References

- Astic, T., and D. W. Oldenburg, 2019, A framework for petrophysically and geologically guided geophysical inversion using a dynamic Gaussian mixture model prior: *Geophysical Journal International*, **219**, no. 3, 1989–2012, <https://doi.org/10.1093/gji/ggz389>.
- Colombo, D., G. McNeice, D. Rovetta, E. Sandoval-Curiel, E. Turkoglu, and A. Sena, 2016, High-resolution velocity modeling by seismic-airborne TEM joint inversion: A new perspective for near-surface characterization: *The Leading Edge*, **35**, no. 11, 977–985, <https://doi.org/10.1190/tle35110977.1>.
- Dadashpour, M., M. Landrø, and J. Kleppe, 2008, Nonlinear inversion for estimating reservoir parameters from time-lapse seismic data: *Journal of Geophysics and Engineering*, **5**, no. 1, 54–66, <https://doi.org/10.1088/1742-2132/5/1/006>.
- Du, Z., and L. M. MacGregor, 2010, Reservoir characterization from joint inversion of marine CSEM and seismic AVA data using genetic algorithms: A case study based on the Luva gas field: 80th Annual International Meeting, SEG, Expanded Abstracts, 737–741, <https://doi.org/10.1190/1.3513888>.
- Gallardo, L. A., and M. A. Meju, 2004, Joint two-dimensional DC resistivity and seismic travel time inversion with cross-gradients constraints: *Journal of Geophysical Research: Solid Earth*, **109**, no. B3, <https://doi.org/10.1029/2003JB002716>.
- Glegola, M., P. Ditmar, R. Hanea, O. Eiken, F. Vossepoel, R. Arts, and R. Klees, 2012, History matching time-lapse surface-gravity and well-pressure data with ensemble smoother for estimating gasfield aquifer support — A 3D numerical study: *SPE Journal*, **17**, no. 4, 966–980, <https://doi.org/10.2118/161483-PA>.
- Hoversten, G. M., F. Cassassuce, E. Gasperikova, G. A. Newman, J. Chen, Y. Rubin, Z. Hou, and D. Vasco, 2006, Direct reservoir parameter estimation using joint inversion of marine seismic AVA and CSEM data: *Geophysics*, **71**, no. 3, C1–C13, <https://doi.org/10.1190/1.2194510>.
- Kamm, J., I. A. Lundin, M. Bastani, M. Sadeghi, and L. B. Pedersen, 2015, Joint inversion of gravity, magnetic, and petrophysical data — A case study from a gabbro intrusion in Boden, Sweden: *Geophysics*, **80**, no. 5, B131–B152, <https://doi.org/10.1190/geo2014-0122.1>.
- Lelièvre, P. G., C. G. Farquharson, and C. A. Hurich, 2012, Joint inversion of seismic traveltimes and gravity data on unstructured grids with application to mineral exploration: *Geophysics*, **77**, no. 1, K1–K15, <https://doi.org/10.1190/geo2011-0154.1>.
- Liang, L., A. Abubakar, and T. M. Habashy, 2016, Reservoir property mapping and monitoring from joint inversion of time-lapse seismic, electromagnetic, and production data: *Geophysics*, **81**, no. 5, WC33–WC44, <https://doi.org/10.1190/geo2015-0620.1>.
- Melo, A. T., J. Sun, and Y. Li, 2017, Geophysical inversions applied to 3D geology characterization of an iron oxide copper-gold deposit in Brazil: *Geophysics*, **82**, no. 5, K1–K13, <https://doi.org/10.1190/geo2016-0490.1>.
- Moorkamp, M., B. Heincke, M. Jegen, A. W. Roberts, and R. W. Hobbs, 2011, A framework for 3-D joint inversion of MT, gravity and seismic refraction data: *Geophysical Journal International*, **184**, no. 1, 477–493, <https://doi.org/10.1111/j.1365-246X.2010.04856.x>.
- Sun, J., and Y. Li, 2016, Joint inversion of multiple geophysical data using guided fuzzy *c*-means clustering: *Geophysics*, **81**, no. 3, ID37–ID57, <https://doi.org/10.1190/geo2015-0457.1>.